

Bank Loan Analysis Query Document

A. Bank Loan Report | Summary

KPI's:

Total Loan Applications:

```
select count(id) as Total_NumberOf_Application from bank_loan_data
```

	Total_NumberOf_Application
1	38576

MTD Total Loan Applications:

```
select count(id) as MTD_Total_Loan_Application from bank_loan_data
```

```
where month(issue_date) = 12 and year(issue_date) = 2021
```

	MTD_Total_Loan_Application
1	4314

PMTD Total Loan Applications:

```
select count(id) as PMTD_Total_Loan_Application from bank_loan_data
```

```
where month(issue_date) = 11 and year(issue_date) = 2021
```

	PMTD_Total_Loan_Application
1	4035

Total Funded Amount:

```
select SUM(loan_amount) as Total_Funded_Amount from bank_loan_data
```

	Total_Funded_Amount
1	435757075

MTD Total Funded Amount:

```
select SUM(loan_amount) as MTD_Total_Funded_Amount from bank_loan_data  
where MONTH(issue_date) = 12 and year(issue_date) = 2021
```

	MTD_Total_Funded_Amount
1	53981425

PMTD Total Funded Amount:

```
select SUM(loan_amount) as PMTD_Total_Funded_Amount from bank_loan_data  
where MONTH(issue_date) = 12 and year(issue_date) = 2021
```

	PMTD_Total_Funded_Amount
1	47754825

Total Amount Received:

```
select SUM(total_payment) AS Total_Amount_Received from bank_loan_data
```

	Total_Amount_Received
1	473070933

MTD Total Amount Received:

```
select SUM(total_amount) as MTD_Total_Amount_Received from bank_loan_data  
where MONTH(issue_date) = 12 and year(issue_date) = 2021
```

	MTD_Total_Amount_Received
1	58074380

PMTD Total Amount Received:

```
select SUM(total_amount) as PMTD_Total_Amount_Received from bank_loan_data  
where MONTH(issue_date) = 11 and year(issue_date) = 2021
```

	PMTD_Total_Amount_Received
1	50132030

Average Interest Rate:

Select ROUND(AVG(int_rate),5) AS Average_interest_Rate from bank_loan_data

	Average_interest_Rate
1	12.04883

MTD Average Interest Rate:

Select ROUND(AVG(int_rate),5) AS MTD_Average_interest_Rate from bank_loan_data
where MONTH(issue_date) = 12 and year(issue_date) = 2021

	MTD_Average_interest_Rate
1	12.35604

PMTD Average Interest Rate:

Select ROUND(AVG(int_rate),5) AS PMTD_Average_interest_Rate from bank_loan_data
where MONTH(issue_date) = 11 and year(issue_date) = 2021

	PMTD_Average_interest_Rate
1	11.94172

Average Debt-to-Income Ratio:

select ROUND(AVG(dti)*100,5) AS Average_Debt_to_Income_Ratio from bank_loan_data

	Average_Debt_to_Income_Ratio
1	13.32743

MTD Average Debt-to-Income Ratio:

select ROUND(AVG(dti)*100,5) AS MTD_Average_Debt_to_Income_Ratio from bank_loan_data
where MONTH(issue_date) = 12 and year(issue_date) = 2021

	MTD_Average_Debt_to_Income_Ratio
1	13.66554

PMTD Average Debt-to-Income Ratio:

```
select ROUND(AVG(dti)*100,5) as PMTD_Average_Debt_to_Income_Ratio from bank_loan_data
where MONTH(issue_date) = 11 and year(issue_date) = 2021
```

	PMTD_Average_Debt_to_Income_Ratio
1	13.30273

Good Loan Percentage:

```
select
(count(case when loan_status = 'Fully Paid' or loan_status = 'current' then id
end)*100)/count(id) as Good_Loan_Percentage
from bank_loan_data
```

	Good_Loan_Percentage
1	86

Good Loan Application:

```
select count(id) as Good_Loan_Application
from bank_loan_data
where loan_status = 'Fully Paid' or loan_status = 'current'
```

	Good_Loan_Application
1	33243

Good Loan Funded Amount:

```
select SUM(loan_amount) as Good_Loan_Funded_Amount
from bank_loan_data
where loan_status = 'Fully Paid' or loan_status = 'current'
```

	Good_Loan_Funded_Amount
1	370224850

Good Loan Total Received Amount:

```
select SUM(total_payment) as Good_Loan_Total_Received_Amount  
from bank_loan_data  
where loan_status = 'Fully Paid' or loan_status = 'current'
```

	Good_Loan_Total_Received_Amount
1	435786170

Bad Loan Percentage:

```
select  
(count(case when loan_status = 'Charged off' or loan_status = 'current' then id  
end)*100)/count(id) as Bad_Loan_Percentage  
from bank_loan_data
```

	Bad_Loan_Percentage
1	13

Good Loan Application:

```
select count(id) as Good_Loan_Application  
from bank_loan_data  
where loan_status = 'Fully Paid' or loan_status = 'current'
```

	Good_Loan_Application
1	5333

Good Loan Funded Amount:

```
select SUM(loan_amount) as Good_Loan_Funded_Amount  
from bank_loan_data  
where loan_status = 'Charged off' or loan_status = 'current'
```

	Good_Loan_Funded_Amount
1	65532225

Bad Loan Total Received Amount:

```
select SUM(total_payment) as Bad_Loan_Total_Received_Amount  
from bank_loan_data  
where loan_status = 'Charged off' or loan_status = 'current'
```

	Bad_Loan_Total_Received_Amount
1	37284763

Loan Status:

```
select  
loan_status,  
count(id) as Total_loan_Application,  
ROUND(AVG(dti)*100,5) AS Average_DTI,  
ROUND(AVG(int_rate)*100,5) AS Average_Interast_Rate,  
sum(loan_amount) as Total_Funded_Amount,  
sum(total_payment) as Total_Received_Amount  
from bank_loan_data  
group by loan_status
```

	loan_status	Total_loan_Application	Average_DTI	Average_Interast_Rate	Total_Funded_Amount	Total_Received_Amount
1	Fully Paid	32145	13.16735	11.64107	351358350	411586256
2	Charged Off	5333	14.00473	13.87857	65532225	37284763
3	Current	1098	14.72434	15.09933	18866500	24199914

Month Analysis:

select

MONTH(issue_date) AS Month_NO,

DATENAME(MONTH, issue_date) as Month_Name,

count(id) as Total_Loan_Application,

sum(loan_amount) as Total_Funded_Amount,

sum(total_payment) as Total_Received_Amount

from bank_loan_data

group by MONTH(issue_date), DATENAME(MONTH, issue_date)

order by MONTH(issue_date)

	Month_NO	Month_Name	Total_Loan_Application	Total_Funded_Amount	Total_Received_Amount
1	1	January	2332	25031650	27578836
2	2	February	2279	24647825	27717745
3	3	March	2627	28875700	32264400
4	4	April	2755	29800800	32495533
5	5	May	2911	31738350	33750523
6	6	June	3184	34161475	36164533
7	7	July	3366	35813900	38827220
8	8	August	3441	38149600	42682218
9	9	September	3536	40907725	43983948
10	10	October	3796	44893800	49399567
11	11	November	4035	47754825	50132030
12	12	December	4314	53981425	58074380

Regional Analysis:

select

address_state,

count(id) as Total_Loan_Application,

sum(loan_amount) as Total_Funded_Amount,

sum(total_payment) as Total_Received_Amount

```

from bank_loan_data

group by address_state

order by address_state

```

	address_state	Total_Loan_Application	Total_Funded_Amount	Total_Received_Amount
1	AK	78	1031800	1108570
2	AL	432	4949225	5492272
3	AR	236	2529700	2777875
4	AZ	833	9206000	10041986
5	CA	6894	78484125	83901234
6	CO	770	8976000	9845810
7	CT	730	8435575	9357612
8	DC	214	2652350	2921854
9	DE	110	1138100	1269136
10	FL	2773	30046125	31601905
11	GA	1355	15480325	16728040
12	HI	170	1850525	2080184
13	IA	5	56450	64482
14	ID	6	59750	65329
15	IL	1486	17124225	18875941
16	IN	9	86225	85521
17	KS	260	2872325	3247394
18	KY	320	3504100	3792530
19	LA	426	4498900	5001160
20	MA	1310	15051000	16676279
21	MD	1027	11911400	12985170
22	ME	3	9200	10808
23	MI	685	7829900	8543660

Loan Term Analysis:

```

select

term,

count(id) as Total_Loan_Application,

sum(loan_amount) as Total_Funded_Amount,

sum(total_payment) as Total_Received_Amount

from bank_loan_data

group by term

```

order by sum(total_payment)

Employee Length Analysis:

select

emp_length,

count(id) as Total_Loan_Application,

sum(loan_amount) as Total_Funded_Amount,

sum(total_payment) as Total_Received_Amount

from bank_loan_data

group by emp_length

order by count(id) desc

	emp_length	Total_Loan_Application	Total_Funded_Amount	Total_Received_Amount
1	10+ years	8870	116115950	125871616
2	< 1 year	4575	44210625	47545011
3	2 years	4382	44967975	49206961
4	3 years	4088	43937850	47551832
5	4 years	3428	37600375	40964850
6	5 years	3273	36973625	40397571
7	1 year	3229	32883125	35498348
8	6 years	2228	25612650	27908658
9	7 years	1772	20811725	22584136
10	8 years	1476	17558950	19025777
11	9 years	1255	15084225	16516173

Loan Purpose Analysis:

select

purpose,

count(id) as Total_Loan_Application,

sum(loan_amount) as Total_Funded_Amount,

sum(total_payment) as Total_Received_Amount

from bank_loan_data

group by purpose

order by count (id) desc

	purpose	Total_Loan_Application	Total_Funded_Amount	Total_Received_Amount
1	Debt consolidation	18214	232459675	253801871
2	credit card	4998	58885175	65214084
3	other	3824	31155750	33289676
4	home improvement	2876	33350775	36380930
5	major purchase	2110	17251600	18676927
6	small business	1776	24123100	23814817
7	car	1497	10223575	11324914
8	wedding	928	9225800	10266856
9	medical	667	5533225	5851372
10	moving	559	3748125	3999899
11	house	366	4824925	5185538
12	vacation	352	1967950	2116738
13	educational	315	2161650	2248380
14	renewable_energy	94	845750	898931

Home OwnerShip Analysis:

select

home_ownership,

count(id) as Total_Loan_Application,

sum(loan_amount) as Total_Funded_Amount,

sum(total_payment) as Total_Received_Amount

from bank_loan_data

group by home_ownership

order by count (id) desc

	home_ownership	Total_Loan_Application	Total_Funded_Amount	Total_Received_Amount
1	RENT	18439	185768475	201823056
2	MORTGAGE	17198	219329150	238474438
3	OWN	2838	29597675	31729129
4	OTHER	98	1044975	1025257
5	NONE	3	16800	19053

